Expanded Concepts — Demo-3 Three-Tier Edition

Long-form essays on the ideas the talk touches but cannot dwell on

DEFCON Coimbatore 2026 · Venugopal Parameswara

> *"The security industry is still selling the same dream it has been selling since the first firewall was deployed. It is a beautiful lie."*

These nine concepts each take about 3 minutes to read. They are the material the 40-minute talk has to compress into a single line.

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1. Why "Graceful Fail" Is Not "Graceful Degradation"

Graceful degradation is the Netflix idea: when the recommendations service is down, show the user a generic list of popular titles. The user's experience is slightly worse, but the service is still serving.

Graceful fail is different. It says: when the container is compromised, *kill it.* Do not try to clean it. Do not try to disinfect it. Do not keep it running while you investigate. Destroy it and respawn from a known-good image. The user experiences a sub-second interruption — and the attacker's foothold disappears.

The distinction matters because "graceful degradation" is a UX concept. Graceful fail is a security concept. One protects the user from backend chaos. The other protects the system from the user.

Demo-3 does both, at different tiers:

- WEB: graceful fail (destroy and respawn — attacker's shell evaporates)
- APP: graceful fail + graceful degradation (sessions lost, but users re-auth)
- DB: neither — the DB is the one thing you cannot destroy. You contain instead. Fence the replica, preserve the clean LSN, accept that some primary rows are now compromised, and failover on the defender's schedule — not the attacker's.

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2. The FAIR Model — Why B Is $5,000 and Not $1,000

FAIR (Factor Analysis of Information Risk) is an ISO/IEC 27005-adjacent quantitative risk framework published by the Open Group. The useful move is not the framework itself — it's the insistence that

risk be expressed as a **probability distribution**, not a point estimate.

For Demo-3's $5,000/min figure, the Monte Carlo simulation runs probability distributions across:

1. **Asset value** — revenue per minute × service dependency (what fraction of revenue runs through this service)
2. **Threat event frequency** — historical breach rate for peer organisations
3. **Vulnerability likelihood** — exploitability of the stack given current controls
4. **Loss form components** — revenue, productivity, response cost, regulatory, reputational

For a Tier-1 service at a $3 billion revenue company, the simulation converges around $5,707/min with a wide band. We round down to $5,000 to keep the number defensible on cross-examination.

Real-world validation:

- Maersk NotPetya: $300M / 14 days ≈ $15,000/min (higher because of supply-chain secondary loss)
- British Airways 2018: ~$53,000/min (higher because of ICO fine, which FAIR's regulatory component captures)
- MGM Resorts 2023 SEC filings: ~$7,000/min (hospitality-sector revenue density)

The FAIR model predicts; the breach data confirms. This is the one security metric that behaves like a real physical quantity.

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3. V = 0 Is Not Hypothetical Anymore

CrowdStrike's 2024 Global Threat Report documented a fastest recorded breakout time of 2 minutes 7 seconds. The average eCrime breakout time is 62 minutes, down from 84 the previous year. The trend is clear and monotonically downward.

AI-assisted exploit generation — not as a theoretical capability but as something already appearing in commodity red-team tooling — compresses the "recon → identify vuln → exploit" phases into seconds. A human-guided attacker using an LLM as an exploit copilot can reach data destruction in well under 30 seconds on an unpatched stack.

What this means for Demo-3:

- **V = 30s is the conservative number for the talk**, not the lower bound
- A fully automated attacker could achieve the same result in under 10 seconds
- The slowest tier's R (Demo-3: DB at 12.5s) sets the defence

If V collapses to 5 seconds in the next 12 months — and that's the trajectory — Demo-3's DB tier loses on that timeline. The answer isn't better detection. The answer is sub-second fence-and-resume, which requires CDC-based monitoring, which is the frontier of this work.

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4. The Immutability Bet

Immutable infrastructure says: never modify a running system. Every change is a new deployment. Every production machine is disposable. If you need to fix something, rebuild it — don't patch it.

This sounds extreme. It isn't. It's how every mature cloud platform works internally — AWS EC2 instances are immutable; you build an AMI and launch. Kubernetes pods are immutable; you build an image and roll out. Terraform encourages immutable infra as the default pattern.

Demo-3's bet: *the same discipline, applied to security.* An attacker who gains container-level access can delete files, inject processes, plant web shells — and none of it survives the next respawn. The attacker's effort has a half-life measured in seconds.

The counter-argument is always: "But what about state?" State is where immutability breaks. Demo-3 takes the argument head-on:

- Web tier is immutable because it has no state (HTML baked at build time)
- App tier is immutable-ish because state has been pushed down to the DB
- DB tier is NOT immutable — and the defence pattern is different (fence-and-resume, not destroy-and-respawn)

The lesson: **push state down until only one tier has it, and defend that tier differently.** This is the single architectural decision that makes B-V-R possible at scale.

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5. Why Guardian Is Bash, Not Kubernetes

A recurring audience question: "Why not use a Kubernetes operator for this?" The honest answer: for the demo, bash is pedagogically correct.

Bash has three properties that matter for the talk:

1. **Readable.** An audience can read the detection logic live. They cannot read a Go operator's reconcile loop live.
2. **Inspectable.** `tail -f /tmp/demo3_guardian.jsonl` is a line-oriented source of truth anyone can grep. Kubernetes events + controller logs + metrics are three separate systems.
3. **Cheap to break.** The CTF asks participants to modify guardian in under an hour. No-one is going to fork and rebuild a Kubernetes operator in an hour.

In production, you should absolutely use a Kubernetes operator. The operator reconcile loop, admission controllers, and PodDisruptionBudget give you the same primitives at scale. Guardian is the teaching version. But the teaching version also has the virtue of not depending on a control plane you don't understand.

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6. The State-Loss Contract

When the APP tier is killed and restarted, session state vanishes. This is not a bug. It's a design decision with a specific name: the **state-loss contract**.

A state-loss contract says: for this tier, when recovery happens, we accept the loss of [specific state X]. Users are told upfront. Engineering is designed for it.

Demo-3's contracts:

Tier	State lost on recovery	Accepted because
WEB	None (no state)	Nothing to lose
APP	In-memory sessions	Users re-authenticate (seconds of friction, not lost data)
DB primary	In-flight transactions (uncommitted)	ACID — uncommitted = rolled back, no corruption
DB replica	Depending on fence timing, ~500 rows during bulk INSERT	Known gap; CDC roadmap closes it

This is different from "best effort" or "we hope nothing breaks." Every piece of state in the system has one of three fates under recovery:

1. **Preserved** (e.g., committed transactions in Postgres)
2. **Accepted loss** (e.g., in-memory sessions)
3. **Known gap** (e.g., the bulk-INSERT leak)

Production systems should document this table for every Tier-1 service. Very few do. The talk argues that this is the single most useful design artefact a security team can produce.

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7. The DB Fence — Why 10 Seconds Is Not Too Slow

R_3 = 12.56s looks slow next to R_1 = 0.63s. Some critics will want it faster. Here's why it shouldn't be — and why "faster" is the wrong goal.

The DB fence has a fundamentally different job from the web respawn:

- **Web respawn:** "This container is broken; throw it away." Cost of being wrong: negligible — worst case, you replaced a good container with a better copy of itself.
- **DB fence:** "The write pattern looks like ransomware. Pause replication." Cost of being wrong: you've halted replication during a legitimate batch job. Users notice. Alerting fires.

The 10 quiet ticks (10 seconds of < 500 rows/sec before resume) are a **statistical filter** to reduce false positives. You don't want to fence, resume, fence, resume, fence — that's worse than just leaving replication alone.

Production refinement paths:

- EWMA + standard deviation instead of a fixed threshold — fences faster on unusual patterns, slower on business-hours batch jobs
- Write-pattern classifier (did this look like a mass UPDATE of a narrow ID range, or a normal bulk INSERT?) — cuts false positives 10×
- Operator-in-the-loop for resume (only auto-fence; human confirms resume) — slower R but zero false resumes

None of those are in the demo. All of them are legitimate upgrades. The 12.56s number is the floor for a reasonable implementation, not the ceiling.

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8. The WAF Is Not The Defence

A specific critique the talk anticipates: "If your WAF blocks 13/15, why do you need any of the other machinery?"

Because the WAF is a layer, not the defence. Consider what each WAF block actually proves:

- Wave 2 (SQLi): the payload never reached the application. **But an application with a SQL injection bug is still vulnerable** — a different payload, a different WAF bypass, a different rule gap, and the WAF misses it. The WAF is a probabilistic filter. The database defence is what matters when the WAF loses a round.
- Wave 3 (XSS): same logic. CSP, output encoding, and HttpOnly cookies are the real defence. The WAF buys time.
- Wave 4 (path traversal): the WAF blocks the obvious patterns, but production path traversal is almost always a combination of partial WAF bypass and a vulnerable file handler.

The three-tier story is the point. If the WAF were the defence, you wouldn't need the respawn layer or the DB fence. The fact that Demo-3 has both — and that they are measurably fast — is the honest architectural answer: no single layer is sufficient. Every layer fails. The defence is the layer under the one that just failed.

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9. What Happens at Layer 4 (Supply Chain)

Demo-3 runs five Docker containers built from five Dockerfiles. The ultimate assumption is that those five images are clean. If they aren't, the entire resilience story falls over.

The three supply-chain failure modes:

1. **Base image poisoning.** The `nginx:alpine`, `python:3.11-alpine`, `postgres:15-alpine` images pull from Docker Hub. If any of those is compromised upstream, everything downstream inherits the compromise. Recent example: the `xz-utils` backdoor was caught in 2024 before it shipped widely, but the attacker had a two-year head start.

2. **Build-time dependency poisoning.** `requirements.txt` names `flask==3.0.3, psycopg2-binary==2.9.9`. A compromised PyPI mirror or a typosquatted package name puts malicious code in the golden image.

3. **CI/CD compromise.** The Dockerfile itself is source code. A compromised git hosting environment or CI runner can modify the Dockerfile between your local commit and the registry push. Nothing in the running container will look wrong.

Counter-measures, in order of increasing difficulty:

1. **cosign verify** on every pulled image — free, 15 minutes to set up
2. **SBOM + Syft/Grype in CI** — free, a day to set up
3. **Signed base images only** (Chainguard Images, Google Distroless) — free, a week to migrate
4. **SLSA Level 3 build provenance** — requires investment in CI hardening
5. **Admission controllers** (Kyverno, OPA Gatekeeper) that reject unsigned images at the cluster edge

CTF Level 4 forces participants to walk through this list. It is the single most important thing you can do in the next 90 days that isn't already in Demo-3.

Closing Thought

The security industry sells walls. This talk sells sprinklers. The wall is necessary — don't stop patching, don't stop blocking, don't stop auditing. But the wall was never going to be enough. It was never going to be enough because no wall is enough, and no-one who sold you one told you that part.

What works is an architecture where *the attacker's success is temporary.* That is not a slogan. That is a measurement. R < V. If you can measure it, you can improve it. If you can improve it, you can survive.

Cyber resilience is not a taller wall. Cyber resilience is a city that can rebuild itself while it is still on fire.

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